

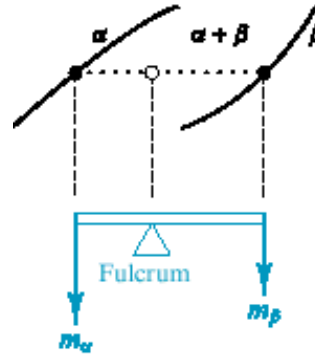


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Metallic Materials and Technology

Mechanical Engineering course

Fe-Fe₃C Metastable equilibrium phase diagram



Lecture n.6

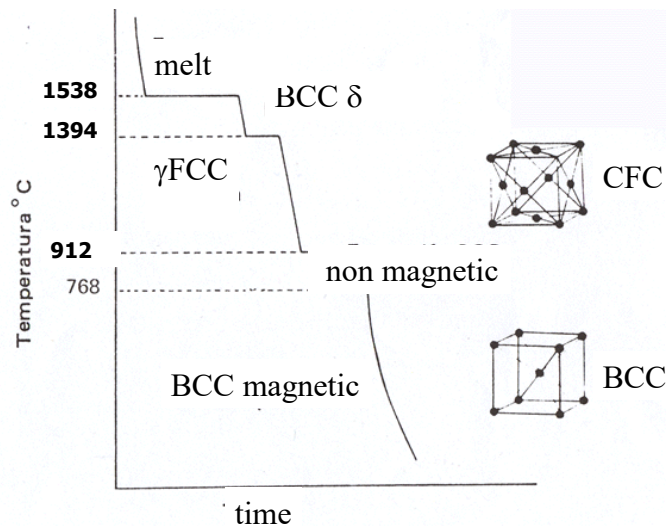
G. Maizza
DISAT Department

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Allotropy of Fe



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Steel Phase Diagram

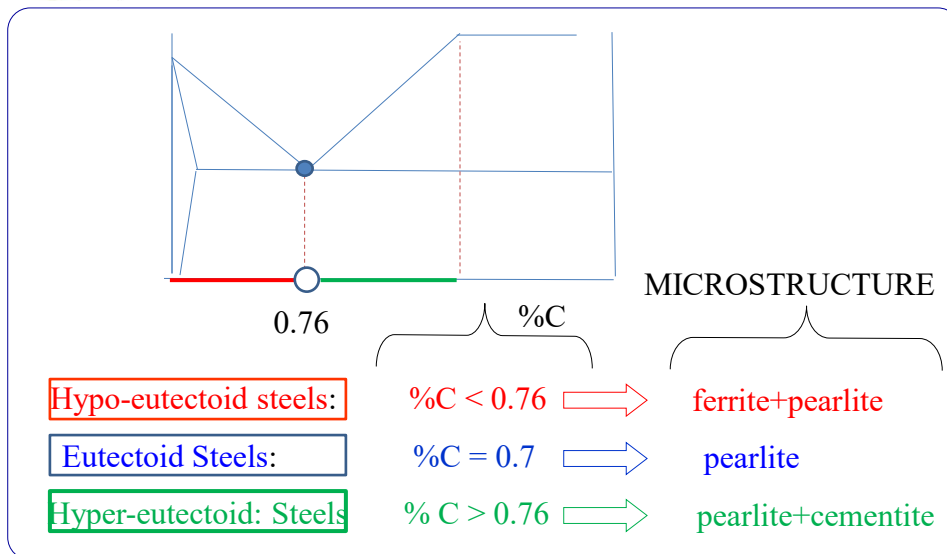
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Classification of Steels as a function of %C



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Steel Phase Diagram

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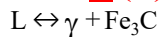


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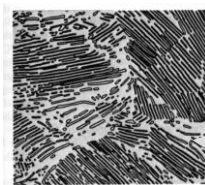
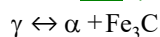
Metastable equilibrium phase Diagram (Fe-Fe₃C)

- two important points

- **Eutectic (A):**

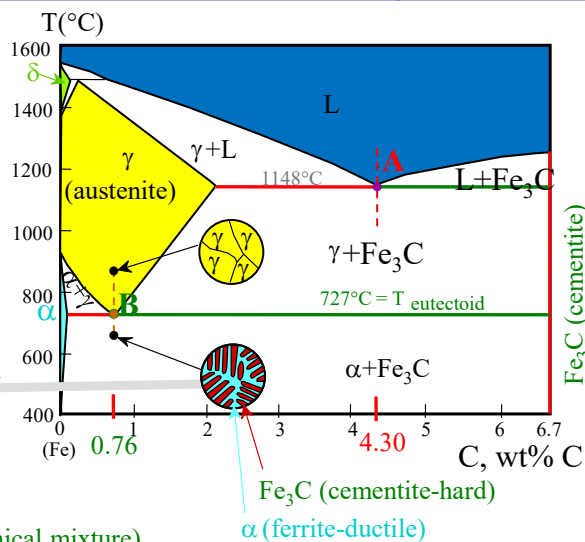


- **Eutectoid (B):**



120 μm

Pearlite =
alternate lamellae of
 α and Fe_3C , (mechanical mixture)



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Steel Phase Diagram

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Invariant points in the Fe-C system

Invariant point	Reaction	phase diagram	location
PERITECTIC	$\text{Liq} + \text{Sol}_1 \xrightleftharpoons[\text{heating}]{\text{cooling}} \text{Sol}_2$		$C=0.16\%C$ $T_P=1493^\circ C$
EUTECTIC	$\text{Liq} \xrightleftharpoons[\text{heating}]{\text{cooling}} \text{Sol}_1 + \text{Sol}_2$		$C=4.3\%$ $T_{EU}=1147^\circ C$
EUTECTOID	$\text{Sol} \xrightleftharpoons[\text{heating}]{\text{cooling}} \text{Sol}_1 + \text{Sol}_2$		$C=0.76\%$ $T_{ED}=727^\circ C$

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Steel Phase Diagram

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EQUILIBRIUM CRITICAL TRANSFORMATION POINTS

A_1 - pearlite/austenite transformation

$C = 0.76\%$, eutectoid temperature

A_3 – austenite/ferrite transformation curve

$C < 0.76\%$, A_3 line

A_{cm} – austenite/cementite transformation line or solubility limit line

of C precipitating as secondary carbide phase

$C > 0.76\%$

Ferrite solubility limit line: solubility line of C in ferrite precipitating as tertiary carbide phase

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NON-EQUILIBRIUM EQUILIBRIUM CRITICAL TRANSFORMATION POINTS

$A_{r1}, (A_{r3})$ Temp. of *end (initial)* formation of ferrite from austenite upon fast cooling

$A_{c1}, (A_{c3})$ Temp. of *initial (end)* formation of austenite from ferrite upon *fast heating*

$B_s, (B_f)$ Temp. of *d'initial (end)* formation of bainite, from austenite, during cooling (*moderately fast cooling*)..

$M_s, (M_f)$ Temp. of *initial (end)* formation of martensite from austenite upon very *fast cooling*

$A_{r\text{cm}}$ Temp. of *initial* formation of cementite from austenite upon slow cooling

$A_{c\text{cm}}$ Temp. of *initial* formation of austenite from cementite upon slow heating

Subscript c denotes “chauffage”, and r denotes «refroidissement»

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INFLUENCE OF ALLOYING ELEMENTS ON TRANSFORMATIONS POINTS

Strong AUSTENITIZING: **Ni, Mn, Co** (a T_{ambient})

AUSTENITIZING: **C, N, Cu**

Strong FERRITIZING: **Cr, Si, Al, Mo, Ti, V,**

Weak FERRITIZING: B, S, Zr, Nb, Ce, Ta

$$A_{c1} = 723 + 16.9 * (1.72 \text{ Si} - 0.63 \text{ Mn} + \text{Cr} - \text{Ni} + 0.37 \text{ W} + 17.2 \text{ As})$$



$$A_{c3} = 910 - 203(\text{C}^{0.5} + 0.15 \text{ Mn} + 0.05 \text{ Cr} + 0.075 \text{ Ni} + 0.1 \text{ Cu} - 3.45 \text{ P} - 0.22 \text{ Si} - 0.16 \text{ Mo} - 0.5 \text{ V} - 0.06 \text{ W} - 0.58 \text{ As} - 1.97 (\text{Al} + \text{Ti}))$$

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Steel Phase Diagram

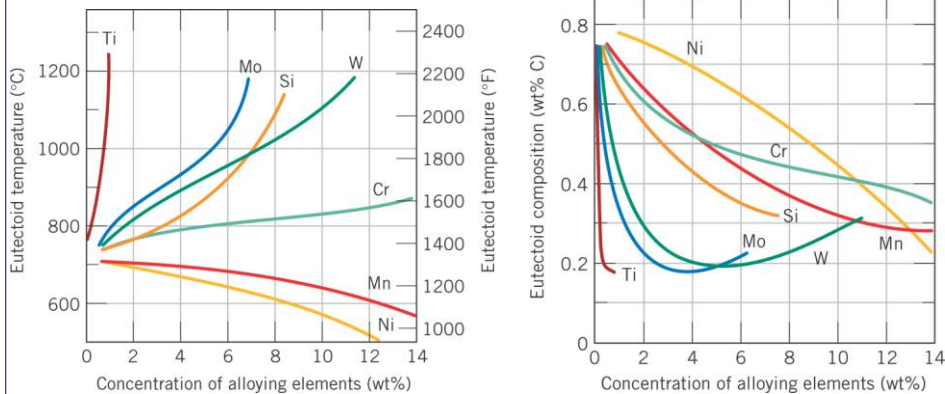
9
Sias, p.566

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INFLUENCE OF ALLOYING ELEMENTS ON A_1 and C_E



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Steel Phase Diagram

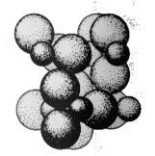
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Cementite



Very hard compound (VHN=1200)
brittle (plastic elong. = 0)

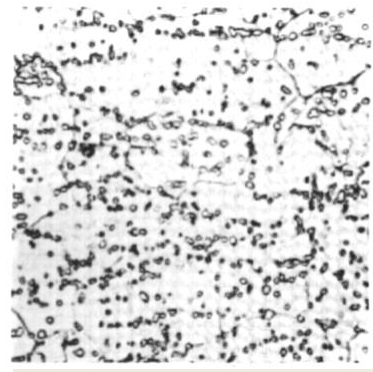
Fe₃C: **orthorhombic Structure**, with **4 at. of C** and **12 of Fe**.

Contains **6.67 w% C** (25% at.)

It is present in all steels when C > 0.025% as (pearlite, bainite) **lamellae**, **needle** (martensite), **spheroids** (after H.T.), or as **film** around austenite grain boundaries (in **iper-eutectoidic steels**).

It markedly affects mechanical strength of steels.

In alloyed steels the iron cementite may be replaced by complex carbides such that Fe → (Cr, Mo, Mn).



Fine spheroidal carbides in ferritic matrix in hyper-eutectoid steels (500 x)

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Steel Phase Diagram

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Sias, p.274, VanVlack, p.118

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Graphite

Graphite has lower density **than** cementite and it is not a metallic phase.

It is promoted by:

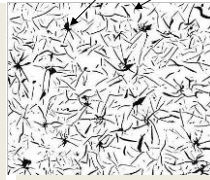
- **extremely slow cooling**
- **high concentration of C** (>2%)
- **+ Si** (1-3 %) (as in cast iron).

In oxydizing environment, graphite has very low friction coefficient, which acts as lubricant in machining operations.

Friction coefficient increases with increasing temperature under both vacuum and controlled atmosphere.

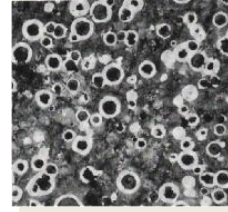
due to inoculating agent

Lamellar graphite



Graphite in gray cast iron. The large black particles are graphite which acts as seed for the new forming graphite.

Strong heat dissipation and vibration/impact damping resistance



Bull-eye type (or spherulitic) cast iron inside a ferrite nodules in a pearlitic matrix.

Strong spheroidizing agents: Mg, Ce.

Optimal combination of mechanical resistance and elongation (18%).

Hi02412/2025, p.118, Non metallurgist, p.49

Steel Phase Diagram

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Ferrite

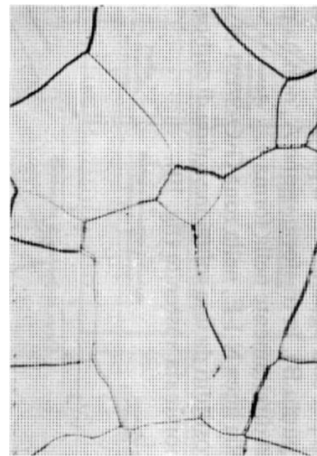
Fe phase; BCC structure.

The most stable at room temperature;

it is often obtained on cooling transformation from austenite

C has very low solubility in this phase; the lowest is at room temperature (**C ≤ 0.0025%**).

Fe atoms can be replaced in the crystal lattice by other alloying elements – e.g. **Cr, Mo, Si**.



Ferrite 500 x

Hi02412/2025

Steel Phase Diagram

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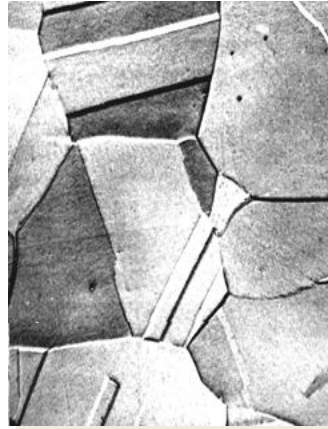
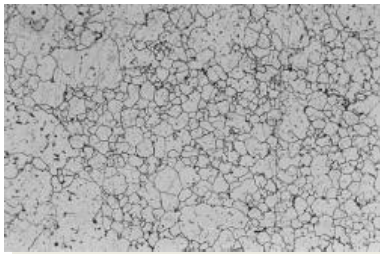


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Austenite

On heating at high temperature the BCC lattice changes to FCC (austenite). When austenite is formed the solubility of any interstitial (e.g. C) into austenite increases

Fe of austenite can be replaced by **Ni** and **Mn**, which easily adapt to the crystal lattice of Fe without significant lattice distortion.



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Steel Phase Diagram

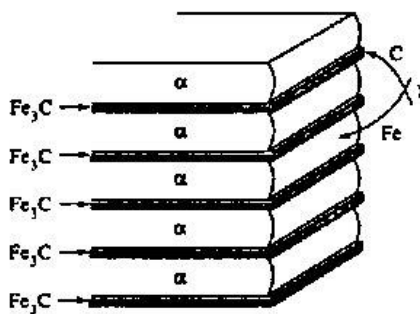
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Growth of pearlite (eutectoid) in steels



On cooling, when austenite reaches A1 temperature, the austenite totally transforms into pearlite with a similar mechanism of eutectic systems

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Steel Phase Diagram

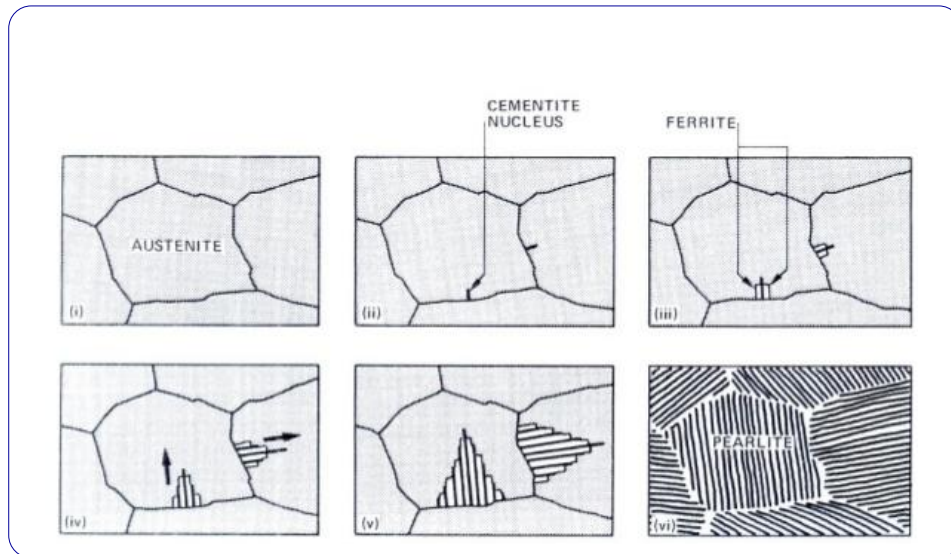
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Nucleation and Growth of pearlite

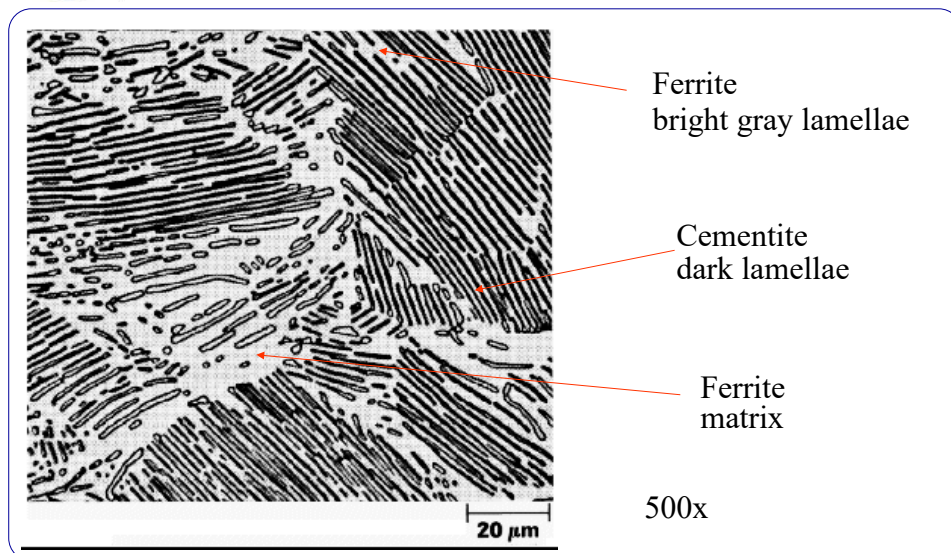


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Pearlite: microstructure



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Steel Phase Diagram

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Pearlite morphology using SEM



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Steel Phase Diagram

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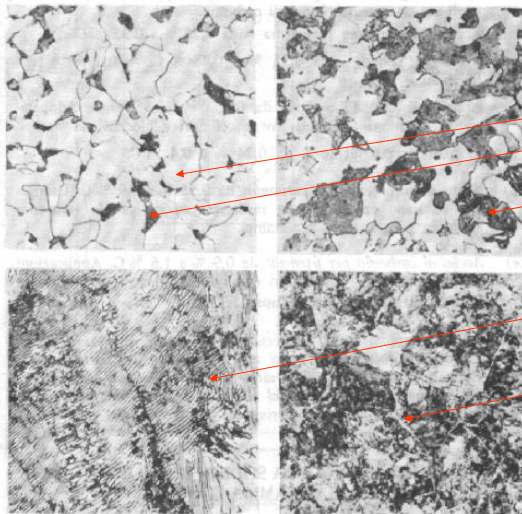
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Plain Carbon steels

hypo-eutectoidic hyper-eutectoidic



Micrographs of plain carbon steels.

a) Low carbon steel with 0.16 % C: ferrite (gray) with small amount of pearlite (dark) (500x);

b) hypo-eutectoid steel with 0.35% C: with increasing amount of pearlite colonies (500x);

c) Near-eutectoid steel 0.85 %C: showing coarse pearlite lamellae (1000x);

d) hyper-eutectoid steel with 1.3 %C: it shows thin intergranular film of cementite (light gray) between pearlite colonies (500x)

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Steel Phase Diagram

Schlenker, p.270

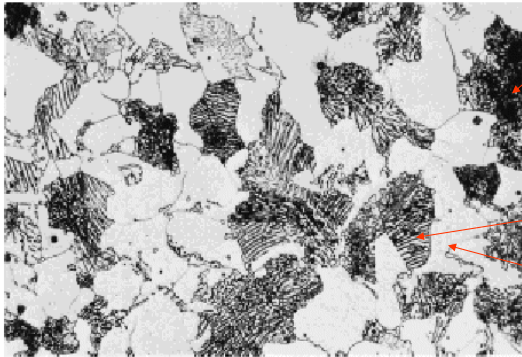
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Hypo-eutectoidic Steel: 0.38%C



Pearlite colony
(dark gray)

535x

pearlite Colonies

pro-eutectoidic Ferrite

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Steel Phase Diagram

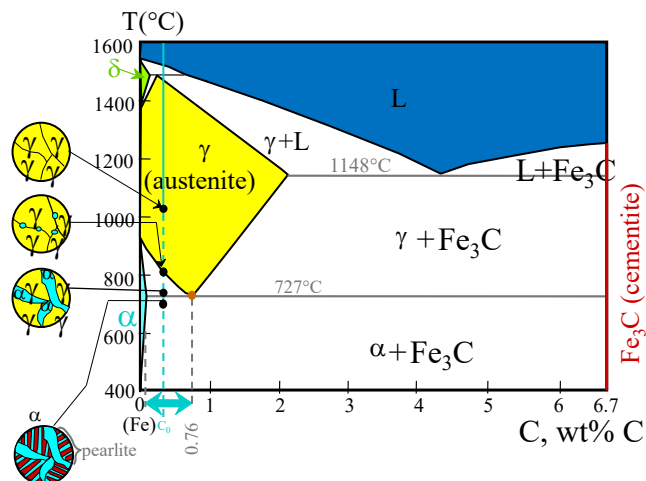
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Transformation on cooling Ipo-eutectoid Steel:



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Steel Phase Diagram

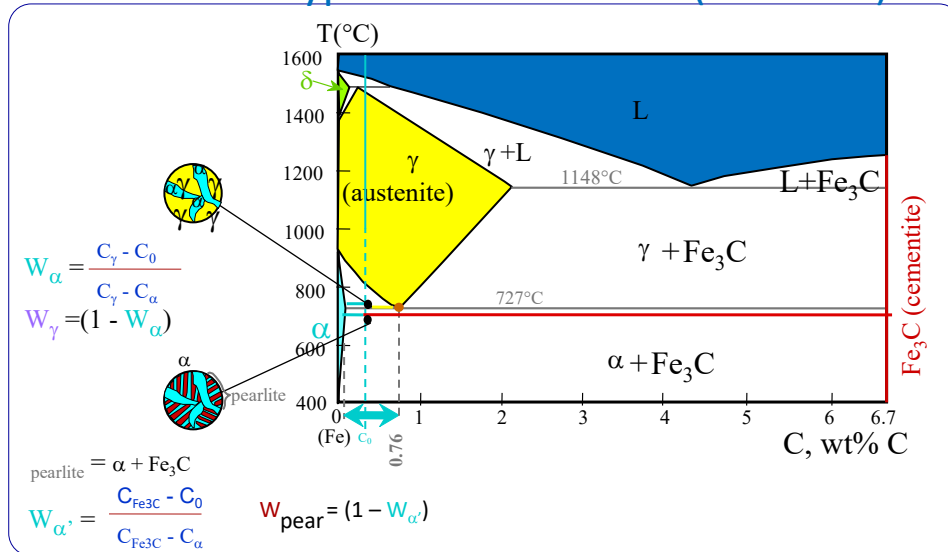
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Lever rule on cooling Hypo-eutectoid Steel ($C < 0.76\%$):



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Steel Phase Diagram

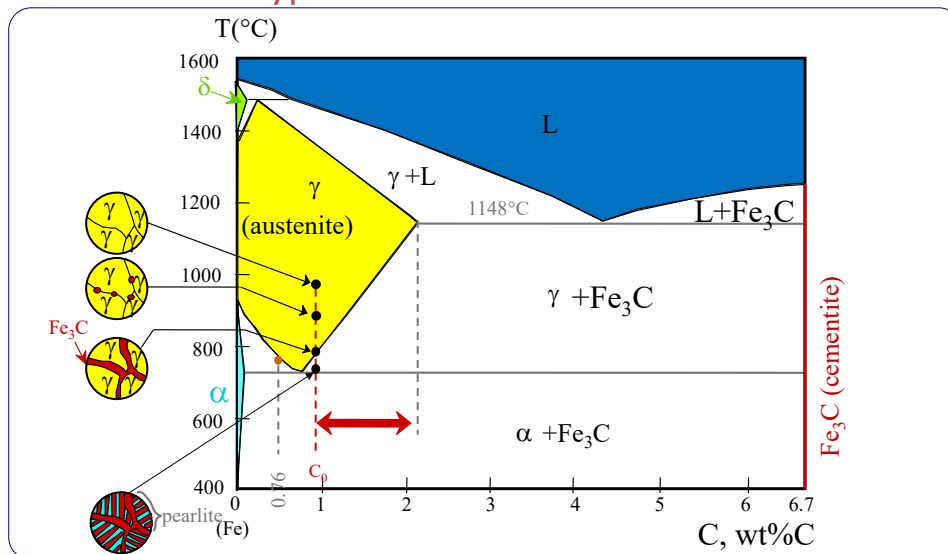
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Transformations on cooling Hyper-eutectoid Steels: $0.76\% < \%C < 2.1$



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Steel Phase Diagram

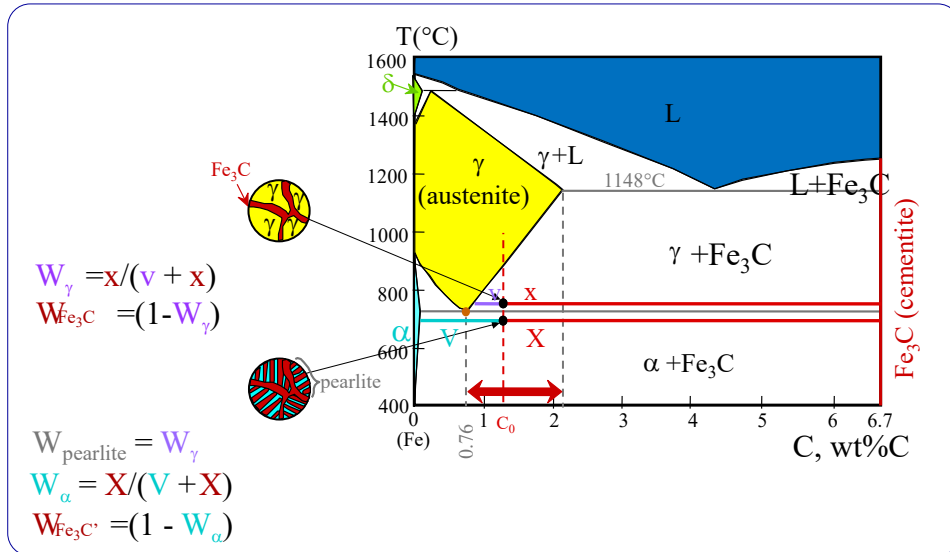
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Lever rule on cooling: Iper-eutectoid Steels: $0.76\% < \%C < 2.1$:



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Steel Phase Diagram

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Cementite in Hyper-eutectoid steels



Pro-eutectoid
cementite which
envelops pearlite
colonies

1000x

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Steel Phase Diagram

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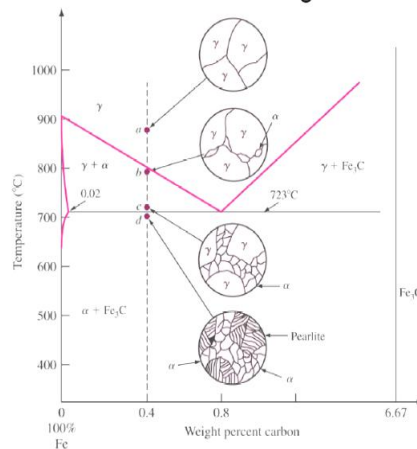
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Exercise

Given an alloy of 99.6 wt% Fe - 0.40 wt% C to a temperature just below A1 point, calculate:

- weight fraction of Fe_3C and ferrite (α).
- the amount in grams of cementite formed per 100 g of alloy.



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Steel Phase Diagram

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Solution

- Use tie line (isotherm) A1

$$C_{\alpha} = 0.022 \text{ wt\% C}$$

$$C_{\text{Fe}_3\text{C}} = 6.70 \text{ wt\% C}$$

- Using the lever rule over the A₁ tie-line

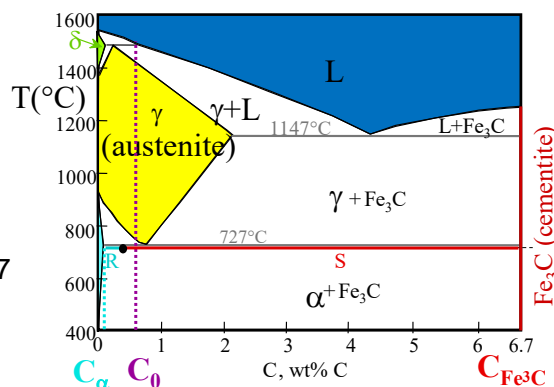
$$W_{\text{Fe}_3\text{C}} = \frac{R}{R+S} = \frac{C_0 - C_{\alpha}}{C_{\text{Fe}_3\text{C}} - C_{\alpha}}$$

$$= \frac{0.40 - 0.022}{6.70 - 0.022} = 0.057$$

Amount of Fe_3C in 100 g

$$= (100 \text{ g}) W_{\text{Fe}_3\text{C}}$$

$$= (100 \text{ g}) (0.057) = 5.7 \text{ g}$$

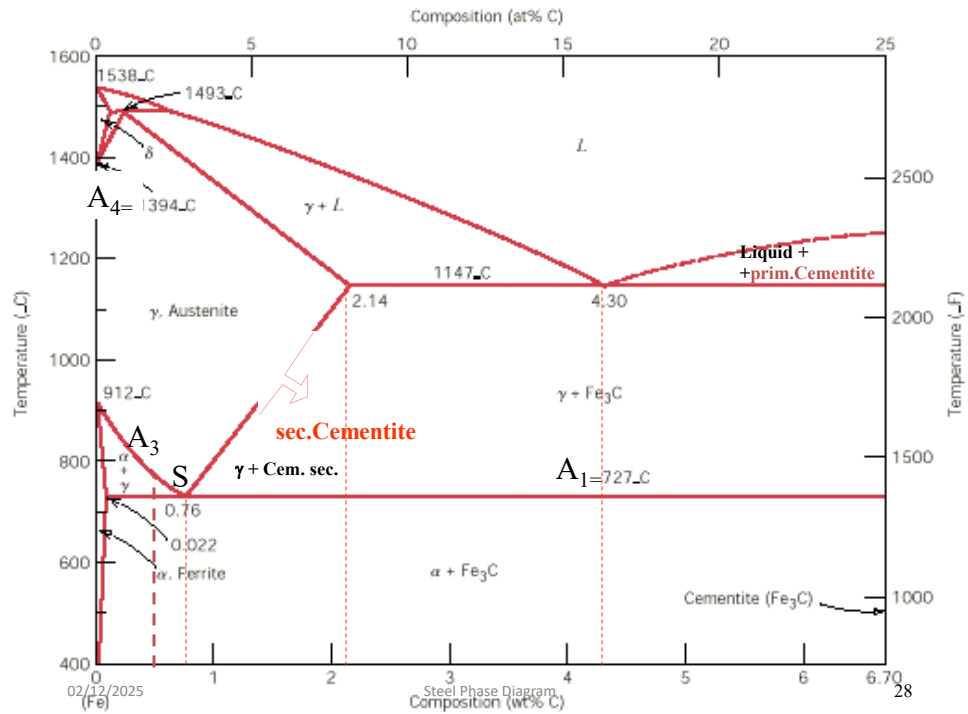


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Steel Phase Diagram

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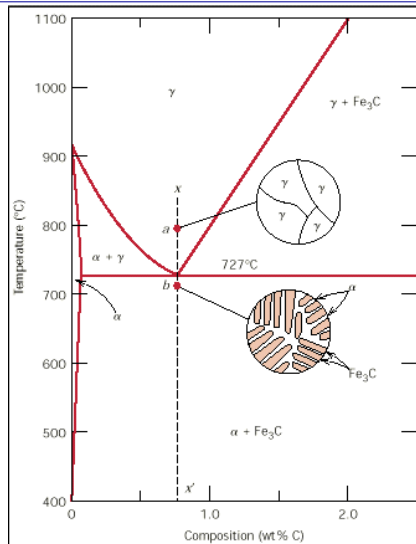
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Eutectoid Steels



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Steel Phase Diagram

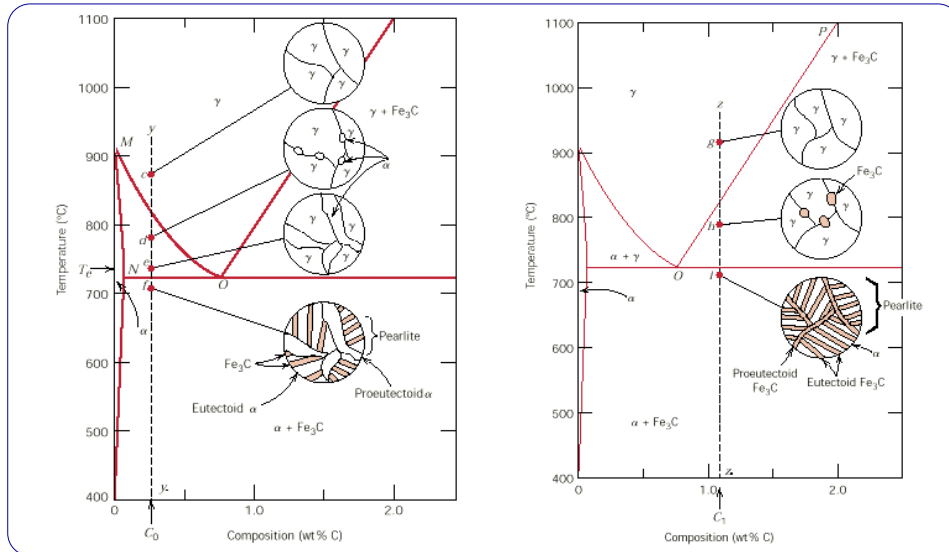
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Hypo- and Hyper-eutectoidic steels



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Steel Phase Diagram

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Self-learning

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Steel Phase Diagram

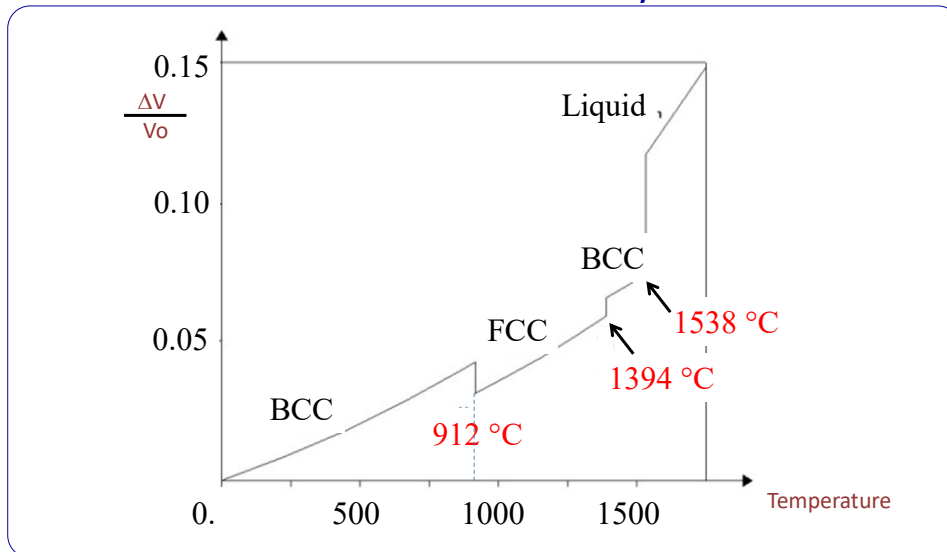
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Polymorphic behaviour of Fe: density

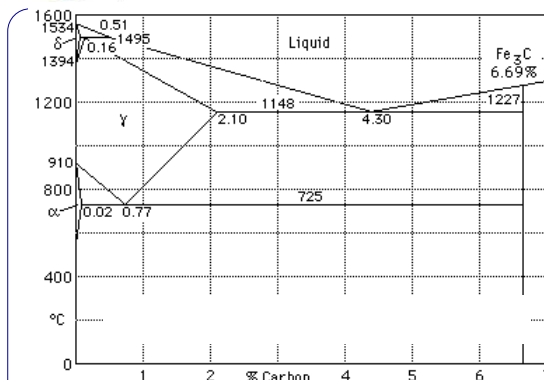


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Weight fraction of Ferrite and cementite in pearlite



Lever rule at T_E^+ :

%wt of ferrite in pearlite:

$$\alpha := \frac{C_{Fe3C} - C_\gamma}{C_{Fe3C} - C_\alpha} \quad \alpha = 88.7\%$$

$$C_\gamma := 0.77\%$$

$$C_\alpha := 0.02\%$$

$$C_{Fe3C} := 6.69\%$$

$$Fe3C := 1 - \alpha$$

$$Fe3C = 11.24\% \quad \text{\%wt of cementite in pearlite:}$$

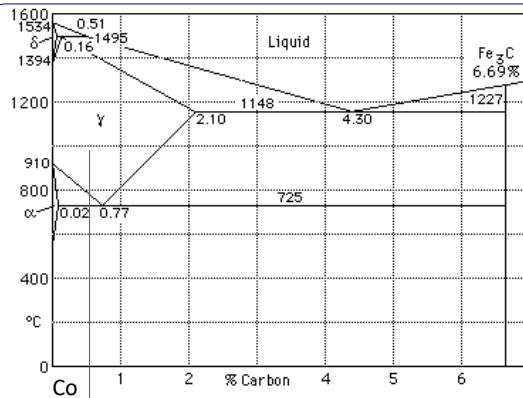
N.B. The calculated fractions in Fe-C phase diagram can be expressed either in %vol or %wt depending on how %C is expressed in the phase diagram: if it is %wt, the phase fractions are also %wt. If the problem asks to determine the thickness of cementite and ferrite lamellae in pearlite, then the respective %vol have to be computed. This is readily done once we know the density of the two phases.

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Lever rule in hypo-eutectoid steels (0.6%C)



Carbon content in main phases:

$$C_{\alpha} := 0.02\%$$

$$C_{\text{Fe}_3\text{C}} := 6.69\%$$

Pro-eutectoid ferrite and cementite in 0.6%C steel:

$$\alpha := \frac{C_{\text{Fe}_3\text{C}} - C_0}{C_{\text{Fe}_3\text{C}} - C_{\alpha}} \quad \alpha = 91.3\% \quad \text{Fe}_3\text{C} = 8.7\%$$

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Lever rule for hypo-eutectoid steels

$$\text{Lever rule at TE}^+: \% \alpha_{\text{proeutectoid}} := \frac{C_{\gamma} - C_0}{C_{\gamma} - C_{\alpha}} \quad C_{\gamma} = 0.77$$

$$\% \alpha_{\text{proeutectoidic}} = 22.67\%$$

$$\% \text{pearlite} = 100 - 22.67 = 77.33\%$$

$$\text{Since: } \% \alpha_{\text{tot}} = 91.3 \Rightarrow$$

$$\% \alpha_{\text{in perl}} = \alpha_{\text{tot}} - \alpha_{\text{pro eutectoid}} = 91.3 - 22.67 = 68.64\%$$

$$\text{Since: } \% \text{wt Cem} = 8.7\% \Rightarrow$$

$$\text{ratio} = \% \alpha / \% \text{Cem (in pearlite)} = 7.89$$

N.B. This ratio stays constant for EACH STEEL; e.g., for pearlite of the 0.77%C steel: ratio = 88.756/11.244 = 7.89.

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Summary of Phase Properties

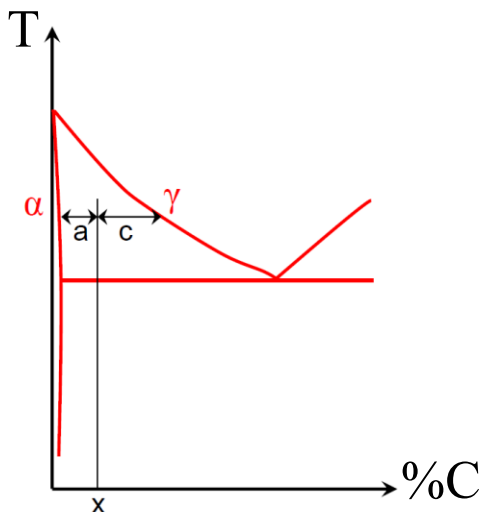
	Rm MPa tensile	HRC/ HB	Crystal structure	ELONG %	Resilience	type
austenite	1034	40 -	FCC	10	high	phase
ferrite	280	0 80	BCC	50	high	phase
cementite	35	- 550	orthorombic a	0	Very low	phase
graphite			hexagonal		Very low	phase
pearlite	800	20 200	Mechanical mixture	10	high	Microconstituent
eutectic	-	-	Mechanical mixture	-	-	Microconstituent

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Fundamentals on Lever Rule



Definitions:

$x = [\%C]$ in the alloy
 $x-a = [\%C]$ in α
 $x+c = [\%C]$ in γ
 f_{α} = weight fraction of α
 f_{γ} = weight fraction of γ

Thus, we obtain:

$$f_{\alpha} (x-a) + f_{\gamma} (x+c) = x$$

$$f_{\alpha} + f_{\gamma} = 1$$

From lever rule:

$$f_{\alpha} = c/(a+c)$$

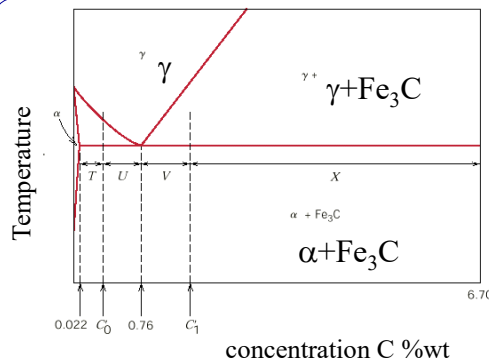
$$f_{\gamma} = a/(a+c)$$

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Lever rule : hypo-eutectoid steels



$C_0' = 0.16\%wt$ Initial %C in the alloy

$$f_p = \frac{T}{T+U} = \frac{C_0' - 0.022}{0.76 - 0.022} = \frac{C_0' - 0.022}{0.74} = 0.186 \quad \text{pearlite (or austenite)}$$

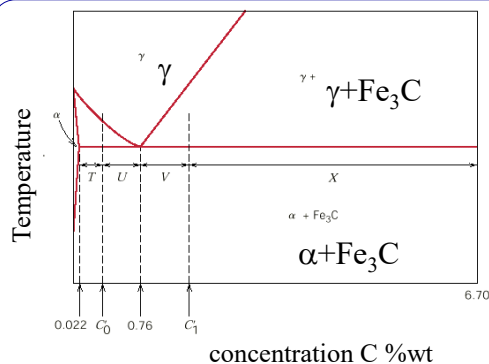
$$f_{\alpha'} = \frac{U}{T+U} = \frac{0.76 - C_0'}{0.76 - 0.022} = \frac{0.76 - C_0'}{0.74} = 0.811 \quad \text{pro-eutectoidic ferrite}$$

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Lever rule : hyper-eutectoid steels



Final results

$$C_1' = 1.3\%wt \quad \begin{cases} f_p = 91\% \\ f_{g'} = 9.1\% \end{cases}$$

$$f_p = \frac{X}{V+X} = \frac{6.67 - C_1'}{6.67 - 0.76} = \frac{6.67 - C_1'}{5.91} = 0.91 \quad \text{Pearlite (or austenite)}$$

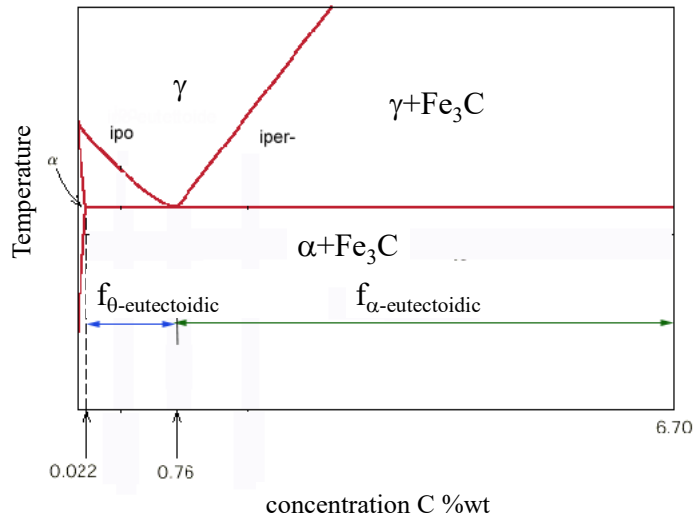
$$f_{g'} = \frac{V}{V+X} = \frac{C_1' - 0.76}{6.67 - 0.76} = \frac{C_1' - 0.76}{5.91} = 0.091 \quad \text{Pro-eutectoidic cementite}$$

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Lever rule: Eutectoid steels

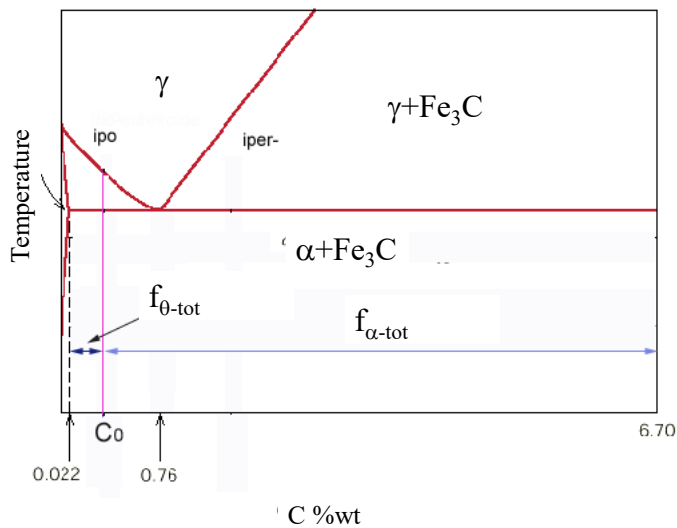


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$f_{\alpha\text{-tot}}$, $f_{\theta\text{-tot}}$: Hypo-eutectoid steels



$$f_{\theta\text{-tot}} = f_{\theta\text{-eutectoidic}}$$

$$f_{\alpha\text{-eutectoidic}} = f_{\alpha\text{-tot}} - f_{\alpha\text{-pro-eutectoidic}}$$

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$f_{\alpha\text{-pro-eutectoidic}}$, f_{pearlite} : hypo-eutectoid steels

